



AIMS 05-01-008

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AIMS
Airbus Material Specification

Carbon Fibre Reinforced Epoxy Prepreg
2,2 Twill fabric/120°C - curing class

Material Specification

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1 Scope

This Material Specification defines the requirements for the 2,2 Twill carbon fiber fabric prepreg, 120°C curing, for aerospace application.

Unless otherwise specified by the drawing, the order, or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

Three grades of prepreg material are covered by this specification:

- Grade A: Prepreg material with low peel behaviour
- Grade B: Grade A prepreg with synthetic fleece (The fleece only in the prepreg face in direct contact with the honeycomb to improve peel behaviour).
- Grade C: Self adhesive prepreg material

2 Application

Manufacture of structural components. Material to be delivered according to the relevant ABS with a theoretical 0,260 mm cured ply thickness for a 42+/-3 % fiber volume fraction .

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 05-01-000 Part 1	Technical Specification - Carbon fibre reinforced epoxy prepreg. General requirements
AIMS 05-01-000 Part 5	Technical Specification - Carbon fiber reinforced epoxy prepreg. Qualification program / Batch release testing. Fabric / 120°C – curing class. Structural material.
AIMS 05-01-000 Part 100	Technical Specification - Carbon fiber reinforced epoxy prepreg Structural material Acceptance Testing and Storage Life Prolongation
ABS 5348	2,2 Twill satin 120°C – curing class. Standard modulus fibre. Structural material
EN 2557	Aerospace series. Carbon fibre preimpregnates. Test method for the determination of mass per unit area.
EN 2558	Aerospace series. Carbon fibre preimpregnates. Test method for the determination of the percentage of volatile matter.
EN 2559	Aerospace series. Carbon fibre preimpregnates. Test method for the determination of resin and fibre content and the mass of fibre per unit area.
EN 2560	Aerospace series. Carbon fibre preimpregnates. Test method for the determination of the resin flow.
EN 2561	Aerospace series. Unidirectional laminates carbon thermosetting resin. Tensile test parallel to the fibre direction.
EN 2563	Aerospace series. Unidirectional laminates carbon thermosetting resin. Test method for the determination of interlaminar shear strength.

EN 2565	Aerospace series. Preparation of carbon fibre reinforced resin panels for test purposes.
EN 2597	Aerospace series. Unidirectional laminates carbon thermosetting resin. Tensile test perpendicular to the fibre direction.
EN 2743	Aerospace series. Standard procedure for conditioning prior to testing.
EN 2823	Aerospace series. Fibre reinforced plastic. Test method for the determination of the effect of exposure to humid atmospheres on physical and mechanical characteristics.
EN 2850	Aerospace series. Fibre reinforced plastic. Testing unidirectional laminates and woven fabric laminates. Compression test.
EN 3615	Aerospace series. Fibre reinforced plastic. Procedure for the determination of the conditions of exposure to humid atmosphere and the determination of moisture absorption.
ISO 1183	Plastics. Methods for determining the density and relative density of non-cellular plastics.
ISO 10119	Carbon fibres. Determination of density.
AITM 1.0002	Aerospace series. Fibre reinforced plastics. Determination of in-plane shear properties ($\pm 45^\circ$ tensile test).
AITM 1.0003	Fibre reinforced plastic materials. Determination of the glass transition temperatures.
AITM 1.0005	Carbon fibre reinforced plastic materials. Determination of interlaminar fracture toughness energy MODE I.
AITM 1.0006	Carbon fibre reinforced plastic materials. Determination of interlaminar fracture toughness energy MODE II.
AITM 1.0007	Fibre reinforced plastics. Determination of notched and unnotched tensile strength.
AITM 1.0008	Fibre reinforced plastics. Determination of notched and unnotched compressive strength of multidirectional plain laminates.
AITM 1.0009	Fibre reinforced plastics. Determination of bearing strength.
AITM 1.0010	Fibre reinforced plastics. Determination of compression strength after impact.
AITM 3.0001	Analysis of thermoset systems by high performance liquid chromatography.
AITM 3.0002	Analysis of non-metallic materials (uncured) by differential scanning calorimetry.
AITM 3.0003	Analysis of organic compounds by infrared spectroscopy.
AITM 3.0004	Determination of the viscosity and gel time of thermosetting resin systems.
ASTM E 831	Standard Test Method for Linear Thermal Expansion of Solid Materials by Thermo-mechanical Analysis.

4 Technical requirements

4.1 General

Specification values are minimum unless otherwise stated.

4.1.1 Worklife and storage conditions

See Table 2.

4.1.2 Processing

See Table 3.

4.2 Specific

4.2.1 Physical-chemical properties

Uncured and cured properties see Table 4.

4.2.2 Mechanical properties

See Table 5 for Grade A and B materials.

See Table 6 for Grade C materials.

5 Designation

Refer to ABS5348.

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS 05-01-000 Part 1.

7 Incoming inspection

Materials qualified against this AIMS shall undergo "Technical Incoming Inspection" as per AIMS 05-01-000 Part 100 prior to use.

Table 1: Material description

Characteristic		Requirements
Material type	Grade A	2,2 Twill carbon fiber fabric/epoxy resin low peel
	Grade B	2,2 Twill carbon fiber fabric/epoxy resin low peel (Grade A) + Synthetic fleece in one face of prepreg
	Grade C	2,2 Twill carbon fiber fabric/epoxy resin self adhesive
Formulation		Fibre: Standard modulus carbon fibre Resin: Epoxy, toughened 120°C cure Fleece: Polyamide fleece 10 g/m ² (for Grade B materials)
Form and method of production		2,2 Twill fabric prepreg, cured in an autoclave at 120°C
Technical Specification		AIMS 05-01-000 Parts 1 and 5
Resin content by weight, uncured		50% nominal

Table 2: Prepreg worklife and storage conditions

No.	Properties	Shop condition	Symbol	Unit	Requirements
1	Storage life at -18°C	-	-	months	Minimum 12
2	Storage life at 5±2°C	-	-	months	-
3	Work life ¹⁾ (tack life)	30°C / 75% R.H.	-	days	-
		25±2°C / 45±10% R.H.	-	days	Minimum 10

¹⁾ The prepreg shall be capable of use after being exposed for an additional minimum of 5 days at the above mentioned shop condition, provided that the lay-up is fully bagged and ready for use.

Table 3: Processing conditions

Cure type	Autoclave		
Cure conditions	Heating rate	°C/min	0,2-5,0
	Cure temperature	°C	115 - 125
	Cure time	min	Max 120
	Cure pressure (nominal)	MPa	0,35
		bar	3,5
	Cooling rate	°C/min	0,2 - 3,5
	Vacuum (abs)	MPa	0,005 - 0,1

Note: The use of an intermediate temperature and low pressure dwell is permissible if agreed between manufacturer and purchaser.

Table 4: Physical properties for CFC fabric prepreg material

No	Properties	Test methods	Unit	Requirements
Material Code				AIMS 05-01-008
				A B C
Uncured				
1	Prepreg areal weight	EN 2557	g/m ²	386 ± 20 (1)
2	Fibre areal weight	EN 2559	g/m ²	193 ± 8 (1)
3	Resin density	ISO 1183 Method A	g/cm ³	1.26 ± 0.04
4	Carbon fibre density	ISO 10119	g/cm ³	1.76 ± 0.04
5	Volatile content	EN 2558	%	< 2
6	Resin content	EN 2559	%	50 ± 3 ¹⁾
7	Physio-chemical tests	AIMS 05-01-000 part 1 (5.3.5.2)	-	See relevant I.P.S.
8	Resin flow	EN 2560	%	See relevant I.P.S.
9	Tack	To be agreed between manufacturer and Airbus	-	See relevant I.P.S.
Cured				
10	Coefficient of thermal expansion	ASTM E831	-	See relevant I.P.S.
11	Water uptake	EN 3615, 70°C/85% R.H./eq.	%	-
12	Water uptake	EN 2378, 70°C/water/eq.	%	See relevant I.P.S.
13	Extent of cure	AITM 3-0008	%	≥ 90
Notes: The stated requirements are average values.				
¹⁾ For Grade B materials these requirements are based taking into account only the carbon fiber fabric / epoxy resin without including the synthetic fleece. The nominal fleece areal weight is 10 g/m ² .				

Table 5: Mechanical properties for Grade A and B materials

Properties	Test Method	Units	Requirements ¹⁾															
			Dry								Wet 70°C/85% R.H. ³⁾						Water 70°C	
			-55°C		RT		70°C		90°C		RT		70°C		80°C		70°C	
			X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin
Glass transition temperature, Tg onset	AITM 1.0003	(°C)	X _{min} = 115								X _{min} = 70						-	
Tensile strength warp	EN 2597 B	(MPa)	450	400	470	430	550	500	-	-	-	-	500	475	-	-	-	-
Tensile modulus warp	EN 2597 B	(GPa)	50± 4		50± 4		50± 4		-		-		50± 4		-		-	
Poisson Ratio	-	-	-	-	0.050±0.005		-	-	-	-	-	-	0.045±0.01		-	-	-	-
Tensile strength weft	EN 2597 B	(MPa)	425	380	460	425	550	525	-	-	-	-	550	525	-	-	-	-
Tensile modulus weft	EN 2597 B	(GPa)	50± 4		50± 4		50± 4		-		-		50± 4		-		-	
Compression strength warp	EN 2850 B	(MPa)	590	550	625	575	535	500	500	450	-	-	480	450	420	375	-	-
Compression modulus warp	EN 2850 B	(GPa)	45± 4		45± 4		45± 4		45± 4		-		45± 4		45± 4		-	
Compression strength weft	EN 2850 B	(MPa)	625	575	580	535	500	470	-	-	-	-	400	370	-	-	-	-
Compression modulus weft	EN 2850 B	(GPa)	45± 4		45± 4		45± 4		-	-	-	-	45± 4		-	-	-	-
In plane shear strength	AITM 1.0002	(MPa)	90	80	95	85	85	75	-	-	-	-	68	65	65	60	-	-
In plane shear modulus	AITM 1.0002	(GPa)	3.60±0. 25		3.20±0. 25		2.50±0. 30		-		-		2.35±0. 25		2.30±0. 25		-	
Compression after impact (2)	AITM 1.0010	(MPa)	-	-	140	125	-	-	-	-	-	-	-	-	-	-	-	-
G _{1C}	AITM 1.0010	(J/m ²)	-	-	425	380	-	-	-	-	500	450	-	-	-	-	-	-
G _{2C}	AITM 1.0006	(J/m ²)	-	-	1750	1600	-	-	-	-	1950	1700	-	-	-	-	-	-
(continued)																		

Table 5: Mechanical properties for Grade A and B materials.

Properties		Test Method	Units	Requirements ¹⁾																	
				Dry								Wet 70°C/85% R.H. ³⁾								Water 70°C	
				-55°C		RT		70°C		90°C		RT		70°C		80°C		70°C			
				X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin		
Plain Tension strength		AITM 1.0007 B (+0/-90) _{2S}	(MPa)	-	-	430	390	-	-	-	-	-	-	390	350	-	-	-	-		
Tension strength (notched) ⁴⁾		AITM 1.0007 B (+0/-90) _{2S}	(MPa)	-	-	180	160	-	-	-	-	-	-	180	160	-	-	-	-		
Filled Hole Tension strength ⁴⁾		AITM 1.0007 A (+0/-90) _{2S}	(MPa)	-	-	180	160	-	-	-	-	-	-	180	160	-	-	-	-		
Plain Compression strength		AITM 1.0007 A (+0/-90) _{2S}	(MPa)	-	-	450	400	-	-	-	-	-	-	310	280	-	-	-	-		
Compression strength (notched) ⁴⁾		AITM 1.0008 B (+0/-90) _{2S}	(MPa)	-	-	250	225	-	-	-	-	-	-	180	160	-	-	-	-		
Filled hole compression strength ⁴⁾		AITM 1.0008 B (+0/-90) _{2S}	(MPa)	-	-	380	350	-	-	-	-	-	-	250	225	-	-	-	-		
Bearing strength	Ult.	AITM 1.0009 B (+0/-90) _{2S}	(MPa)	-	-	675	600	-	-	-	-	-	-	575	520	500	475	-	-		
	2% elong.			-	-	450	400	-	-	-	-	-	-	320	290	310	280	-	-		
Interlaminar shear strength warp		EN 2563	(MPa	60	54	60	58	50	45	50	4-5	-	-	33	29	29	26	-	-		
Interlaminar shear strength (reference)		EN 2563	(MPa)	-	-	60	58	-	-	-	-	-	-	33	29	-	-	-	-		
Interlaminar shear strength (skydrol, 70°C/1000 h)		EN 2563	(MPa)	-	-	-	-	⁶⁾		-	-	-	-	-	-	-	-	-	-		
Interlaminar shear strength (skydrol/water, 50°C/1000 h)		EN 2563	(MPa)	-	-	-	-	⁶⁾		-	-	-	-	-	-	-	-	-	-		
Interlaminar shear strength (fuel, RT/1000 h)		EN 2563	(MPa)	-	-	-	-	⁶⁾		-	-	-	-	-	-	-	-	-	-		
Interlaminar shear strength (MEK, RT/1 h)		EN 2563	(MPa)	-	-	⁶⁾		-	-	-	-	-	-	-	-	-	-	-	-		
(continued)																					

Table 5: Mechanical properties for Grade A and B materials (concluded)

Properties	Test Method	Units	Requirements ¹⁾															
			Dry								Wet 70°C/85% R.H. ³⁾						Water 70°C	
			-55°C		RT		70°C		90°C		RT		70°C		80°C		70°C	
			X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin
In plane shear strength (phenolic paint stripper)	AITM 1.0001	(MPa)	-	-	6)		-	-	-	-	-	-	-	-	-	-	-	
In plane shear strength (non-phenolic paint stripper)	AITM 1.0001	(MPa)	-	-	6)		-	-	-	-	-	-	-	-	-	-	-	
Flatwise tensile strength ⁵⁾	AITM 1.0025	(Mpa)	4.0	3.8	4.0	3.8	4.0	3.8	4.0	3.8	-	-	2.0	1.8	-	-	-	
Fluid tightness test ⁵⁾	AITM 2.0037	-	-	-	Pass		-	-	-	-	-	-	-	-	-	-	-	
Climbing drum peel ⁵⁾	EN 2243-3	N m/m	-	-	30	25	-	-	-	-	-	-	-	-	-	-	-	

¹⁾ All requirements calculated using a nominal thickness per ply of 0.260 mm except Interlaminar Shear, In Plane Shear, Gic, Bearing and sandwich properties in which actual thickness was used.

- Xmeans minimum average value.
- Xmin means minimum individual value.
- The specimens for mechanical testing were manufactured using Grade A material (material without fleece) except tests: Flatwise tensile strength, Fluid Tightness and Climbing Drum Peel tests in which, for the plies in direct contact with the honeycomb core, the material with synthetic fleece (Grade B) was used.

²⁾ Lay-up (+/0/-/90)2S. Impact energy 25 Joules.

³⁾ According to EN 2823.

⁴⁾ Lay-up (+/0/-/90)2S . Requirements using the total width of the specimens, without discounting the hole diameter

⁵⁾ For Climbing drum peel tests: Lay up +45°/-45°/ Nomex HC 4.0-1/8 / -45° /+45

- For Flatwise tensile tests: Lay up 0° / 90° / Glass HC 4.0-3 /16 / 90° / 0°
- For Fluid tightness tests lay up to be defined during qualification

⁶⁾ Values after conditioning will not have a decrease higher that the corresponding one in comparison with the wet reference value tested at equivalent temperature

Table 6: Mechanical properties for Grade C materials

Properties	Test Method	Units	Requirements ¹⁾															
			Dry								Wet 70°C/85% R.H. ⁴⁾						Water 70°C	
			-55°C		RT		70°C		90°C		RT		70°C		80°C		70°C	
			X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin
Glass transition temperature, Tg onset	AITM 1.0003	(°C)	X _{min} = 110								X _{min} = 70						-	
Tensile strength warp	EN 2597 B	(MPa)	500	450	625	560	590	530	-	-	-	-	560	515	-	-	-	-
Tensile modulus warp	EN 2597 B	(GPa)	52 ± 6		52 ± 6		52 ± 6		-	-	-	-	52 ± 6		-	-	-	-
Poisson Ratio	-	-	-	-	0.050 ± 0.005		-	-	-	-	-	-	0.045 ± 0.01		-	-	-	-
Tensile strength weft	EN 2597 B	(MPa)	500	450	600	550	570	525	-	-	-	-	560	515	-	-	-	-
Tensile modulus weft	EN 2597 B	(GPa)	52 ± 6		52 ± 6		52 ± 6		-	-	-	-	52 ± 6		-	-	-	-
Compression strength warp	EN 2850 B	(MPa)	600	540	535	500	430	380	375	340	-	-	310	280	105	95	-	-
Compression modulus warp	EN 2850 B	(GPa)	46± 6		46± 6		46± 6		46± 6		-	-	46± 6		46± 6		-	-
Compression strength weft	EN 2850 B	(MPa)	550	495	535	495	430	380	-	-	-	-	250	230	105	95	-	-
Compression modulus weft	EN 2850 B	(GPa)	46± 6		46± 6		46± 6		-	-	-	-	46± 6		46± 6		-	-
In plane shear strength	AITM 1.0002	(MPa)	100	95	95	90	80	75	-	-	-	-	60	50	58	48	-	-
In plane shear modulus ²⁾	AITM 1.0002	(GPa)	4.50±0. 35		3.65±0. 35		3.50±0. 35		-	-	-	-	1.25±0. 35		1.0±0. 35		-	-
Compression after impact ³⁾	AITM 1.0010	(MPa)	-	-	180	175	-	-	-	-	-	-	-	-	-	-	-	-
G _{1C}	AITM 1.0010	(J/m ²)	-	-	675	600	-	-	-	-	700	650	-	-	-	-	-	-
G _{2C}	AITM 1.0006	(J/m ²)	-	-	2200	1900	-	-	-	-	2200	1900	-	-	-	-	-	-
(continued)																		

Table 6: Mechanical properties for Grade C materials.

Properties		Test Method	Units	Requirements ¹⁾																	
				Dry								Wet 70°C/85% R.H. ³⁾								Water 70°C	
				-55°C		RT		70°C		90°C		RT		70°C		80°C		70°C			
				X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin		
Plain Tension strength		AITM 1.0007 B (+0/-90) _{2S}	(MPa)	-	-	450	400	-	-	-	-	-	-	400	360	-	-	-	-		
Tension strength (notched) ⁵⁾		AITM 1.0007 B (+0/-90) _{2S}	(MPa)	-	-	200	180	-	-	-	-	-	-	200	180	-	-	-	-		
Filled Hole Tension strength ⁵⁾		AITM 1.0007 A (+0/-90) _{2S}	(MPa)	-	-	180	160	-	-	-	-	-	-	180	160	-	-	-	-		
Plain Compression strength		AITM 1.0007 A (+0/-90) _{2S}	(MPa)	-	-	400	360	-	-	-	-	-	-	270	240	-	-	-	-		
Compression strength (notched) ⁵⁾		AITM 1.0008 B (+0/-90) _{2S}	(MPa)	-	-	200	180	-	-	-	-	-	-	140	125	-	-	-	-		
Filled hole compression strength ⁵⁾		AITM 1.0008 B (+0/-90) _{2S}	(MPa)	-	-	250	225	-	-	-	-	-	-	160	140	-	-	-	-		
Bearing strength	Ult.	AITM 1.0009 B (+0/-90) _{2S}	(MPa)	-	-	690	660	-	-	-	-	-	-	430	390	325	300	-	-		
	2% elong.			-	-	560	530	-	-	-	-	-	-	425	370	275	250	-	-		
Interlaminar shear strength warp		EN 2563	(MPa)	65	59	60	55	40	35	26	24	-	-	19	17	15	14	-	-		
Interlaminar shear strength (reference)		EN 2563	(MPa)	-	-	60	55	-	-	-	-	-	-	-	-	-	-	-	-		
Interlaminar shear strength (skydrol, 70°C/1000 h)		EN 2563	(MPa)	-	-	-	-	7)	7)	-	-	-	-	-	-	-	-	-	-		
Interlaminar shear strength (skydrol/water, 50°C/1000 h)		EN 2563	(MPa)	-	-	-	-	7)	7)	-	-	-	-	-	-	-	-	-	-		
Interlaminar shear strength (fuel, RT/1000 h)		EN 2563	(MPa)	-	-	-	-	7)	7)	-	-	-	-	-	-	-	-	-	-		
Interlaminar shear strength (MEK, RT/1 h)		EN 2563	(MPa)	-	-	7)	7)	-	-	-	-	-	-	-	-	-	-	-	-		
(continued)																					

Table 6: Mechanical properties for Grade C materials. (concluded)

Properties	Test Method	Units	Requirements ¹⁾																	
			Dry								Wet 70°C/85% R.H. ³⁾								Water 70°C	
			-55°C		RT		70°C		90°C		RT		70°C		80°C		70°C			
			X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin	X	Xmin		
In plane shear strength (phenolic paint stripper)	AITM 1.0001	(MPa)	-	-	7)	7)	-	-	-	-	-	-	-	-	-	-	-	-		
In plane shear strength (non-phenolic paint stripper)	AITM 1.0001	(MPa)	-	-	7)	7)	-	-	-	-	-	-	-	-	-	-	-	-		
Flatwise tensile strength ⁶⁾	AITM 1.0025	(Mpa)	5.0	4.0	5.0	4.0	5.0	4.0	5.0	4.0	-	-	3.0	2.5	-	-	-	-		
Fluid tightness test ⁶⁾	AITM 2.0037	-	-	-	Pass		-	-	-	-	-	-	-	-	-	-	-	-		
Climbing drum peel ⁶⁾	EN 2243-3	N m/m	-	-	30	25	-	-	-	-	-	-	-	-	-	-	-	-		

¹⁾ All requirements calculated using a nominal thickness per ply of 0.260 mm except Interlaminar Shear, In Plane Shear, Gic, Bearing and sandwich properties in which actual thickness was used.

- Xmeans minimum average value.
- Xmin means minimum individual value.
- The specimens for mechanical testing were manufactured using Grade A material (material without fleece) except tests: Flatwise tensile strength, Fluid Tightness and Climbing Drum Peel tests in which, for the plies in direct contact with the honeycomb core, the material with synthetic fleece (Grade B) was used.

²⁾ Requirements to be reviewed

³⁾ Lay-up (+/0/-90)2S. Impact energy 25 Joules.

⁴⁾ According to EN 2823.

⁵⁾ Lay-up (+/0/-90)2S . Requirements using the total width of the specimens, without discounting the hole diameter

⁶⁾ For Climbing drum peel tests: Lay up +45°/-45°/ Nomex HC 4.0-1/8 / -45° /+45

- For Flatwise tensile tests: Lay up 0° / 90° / Glass HC 4.0-3 /16 / 90° / 0°
- For Fluid tightness tests lay up to be defined during qualification

⁷⁾ Values after conditioning will not have a decrease higher that the corresponding one in comparison with the wet reference value tested at equivalent temperature

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 10/03		New Standard
2 06/04	4	General Revision
3 04/15	2	AIMS05-01-000 Part 100 added
	7	Chapter 7 defined in line with incoming inspection (Part 100) implementation